Assignment 4. Preparation for the defense quiz

The defense of Assignment 4 is a quiz in Moodle. This file tells you what it contains. The questions in the quiz use other numbers and other words than the examples here, but they are of exactly these types.

1. Format

- 20 questions, 30 minutes, 50 points (2.5 points per question). One attempt.
- 3 easy warm-up questions, 3 small calculations, 14 theory questions.
- Question types: choose one answer of four; select all the right answers (each wrong choice lowers the mark, the minimum is 0); match the lines with answers from drop-down lists; true or false; type a whole number.
- Paper and a pen are allowed. Phones, messages and AI tools are not allowed.
- Every student gets a different set of questions from a bigger bank.
- Every calculation question explains the task in the text (what a prefix function is, what the edit distance is), so you do not have to remember a definition word by word. You have to know how the method works.

How to prepare: every calculation and every theory topic below is shown step by step in the animations of your page output/outbreak.html. Repeat each one by hand on the tiny example from examples/tiny, then on a word of your own.

2. Easy warm-up and calculations

2.0 Easy warm-up (3 questions)

Very short questions that check that you have read the assignment. Examples: how many links has a tree of 12 people (11)? How many cells has the edit distance table of a word with 5 letters and a word with 7 letters (6 * 8 = 48)? Which letters has a DNA word (A, C, G, T)? What is the hash of a window in Rabin-Karp (a number made from its letters)?

The three calculations are of these kinds. The words are short (3 to 6 letters), the trees have 5 or 6 people. Each takes 1 to 3 minutes by hand. Here is one worked example of each kind.

2.1 Edit distance or LCS of two short words (1 question)

Task: A = ACTG, B = ATG. Insertion, deletion and replacement cost 1. Edit distance? Longest common subsequence?

```
edit distance table (row = prefix of A, column = prefix of B)
     -  A  T  G
 -    0   1   2   3
 A    1   0   1   2
 C    2   1   1   2
 T    3   2   1   2
 G    4   3   2   1

LCS table
     -  A  T  G
 -    0   0   0   0
 A    0   1   1   1
 C    0   1   1   1
 T    0   1   2   2
 G    0   1   2   3
```

Edit distance 1 (delete C). LCS 3 (ATG). Rule of the first table: d[i][j] = the smallest of d[i-1][j-1] (+1 if the letters differ), d[i-1][j] + 1, d[i][j-1] + 1. Rule of the second: equal letters take d[i-1][j-1] + 1, different letters take the larger of the two neighbours.

2.2 Prefix function or number of occurrences (1 question)

Task: the marker is ACACGAC. What is the last value of the prefix function (the Border length)?

```
i      0 1 2 3 4 5 6
letter A C A C G A C
pi     0 0 1 2 0 1 2
```

The Border length is 2: AC is both the beginning and the ending of the whole word. Another task of this kind: how many times does CGC occur in CGCGCG, occurrences may overlap? Answer 2 (positions 0 and 2).

2.3 DP on a small tree (1 question)

Task: 6 people, links 0-1, 0-2, 2-3, 2-4, 4-5, costs of testing 4, 2, 3, 1, 5, 2. Every link needs a tested end. Smallest total cost? Longest chain in links?

```
person  0  1  2  3  4  5
tested  9  2  5  1  5  2
not     7  0  6  0  2  0
(go from the leaves to the root: tested = cost + sum of min(child tested, child not);
 not tested = sum of child tested)
answer = min(9, 7) = 7   (test persons 1, 2 and 5: 2 + 3 + 2)
longest chain: 1 - 0 - 2 - 4 - 5, that is 4 links
```

3. Theory (14 questions): what you must be able to explain

Topics and the number of questions: dynamic programming, edit distance and LCS (4), prefix function and KMP (4), Rabin-Karp (3), trees and the DP on a tree (3). For every topic you should be able to answer the questions below in one or two sentences, and choose the right sentence among four.

Dynamic programming, edit distance, LCS

- What edit distance measures, and how it differs from comparing letter by letter (Hamming distance).
- The three options that the minimum is taken from in a cell, and the meaning of row 0 and column 0.
- The difference between a subsequence and a substring. The rule of the LCS table when the letters are equal and when they differ.
- Time and memory of the table method, O(n*m), and when two rows are enough.
- When DP applies: smaller subproblems of the same kind, and the same subproblems repeat.

Prefix function and KMP

- What pi[i] means; what the Border length of a marker means; why the pointer in the text never goes back.
- Why KMP is O(n + m) and the simple search is O(n*m) in the worst case; an example of the worst case.
- What to do after a full match (k = pi[m-1]) and why. Why the prefix function is computed in O(m).

Rabin-Karp

- The idea of the rolling hash and why its update costs O(1).
- Equal hashes do not prove equal words (compare the letters); different hashes prove different words. Why a modulus is used.
- Average time O(n) and the worst case O(n*k); how it finds many equal-length words with a set of hashes.

Trees and the DP on a tree

- A tree with n vertices has n-1 links and exactly one path between two vertices.
- The two states of a person (tested and not tested); if a person is not tested, every child must be tested.
- The order of the computation (from the leaves to the root), the time O(n), and why the greedy choice does not always work.
- The longest chain through a person: the two deepest branches below him.

4. Glossary of words used in the quiz

- prefix function: the array pi of a word. In the lectures it is called the LPS array (longest proper prefix that is also a suffix).
- prefix: the beginning of a word. Example: AC and ACG are prefixes of ACGT. suffix: the end of a word. Example: GT and CGT are suffixes of ACGT.
- substring: a piece of a word with no gaps (CG in ACGT). subsequence: letters in the same order, gaps allowed (AT in ACGT).
- border: a piece that is both the beginning and the ending of a word, and is shorter than the word. Example: AC in ACGAC.
- DP (dynamic programming): split a problem into smaller ones, save their answers in a table, use them again. subtree: a vertex and everything below it. vertex = node = person; edge = link.
- occurrence: one place where a word appears in a text. overlap: two occurrences that share letters.

- hash: a number made from a word. Equal words give equal hashes. collision: two different words with the same hash. window: a piece of the text of fixed length that moves along the text.

5. Practice questions

1. The edit distance of ACGT and AGT is: (a) 0 (b) 1 (c) 2 (d) 3

2. The LCS of ACGT and AGT has length: (a) 2 (b) 3 (c) 4 (d) 1

3. The last value of the prefix function of ACACA is: (a) 1 (b) 2 (c) 3 (d) 0

4. Which statement about KMP is right? (a) the pointer in the text goes back after a mismatch (b) the pointer in the text never goes back (c) it sorts the pattern first (d) it works only for words of 4 letters

5. Rabin-Karp: two windows have equal hashes. What do you do? (a) count them as equal (b) skip them (c) compare the letters (d) change the hash

6. A tree has 9 people. How many links does it have? (a) 8 (b) 9 (c) 10 (d) 7

7. The time of the table method for the edit distance of words of length n and m is: (a) O(n+m) (b) O(n*m) (c) O(n^2*m^2) (d) O(log n)

8. In the tree DP, a person is not tested. Then his children are: (a) all tested (b) all not tested (c) one tested (d) free to choose

9. Select all the right answers. Which statements about KMP are true? (a) the pointer in the text never goes back (b) after a mismatch k becomes pi[k-1] (c) the text must be sorted (d) the pattern is read only once and the prefix function is not needed

10. Match each word with its meaning: hash, collision, window, prefix. Meanings: a number made from a word; two different words with the same hash; a piece of the text with a fixed length; the beginning of a word.

Answers: 1: (b); 2: (b); 3: (c); 4: (b); 5: (c); 6: (a); 7: (b); 8: (a); 9: (a) and (b); 10: hash = a number made from a word, collision = two different words with the same hash, window = a piece of the text with a fixed length, prefix = the beginning of a word..